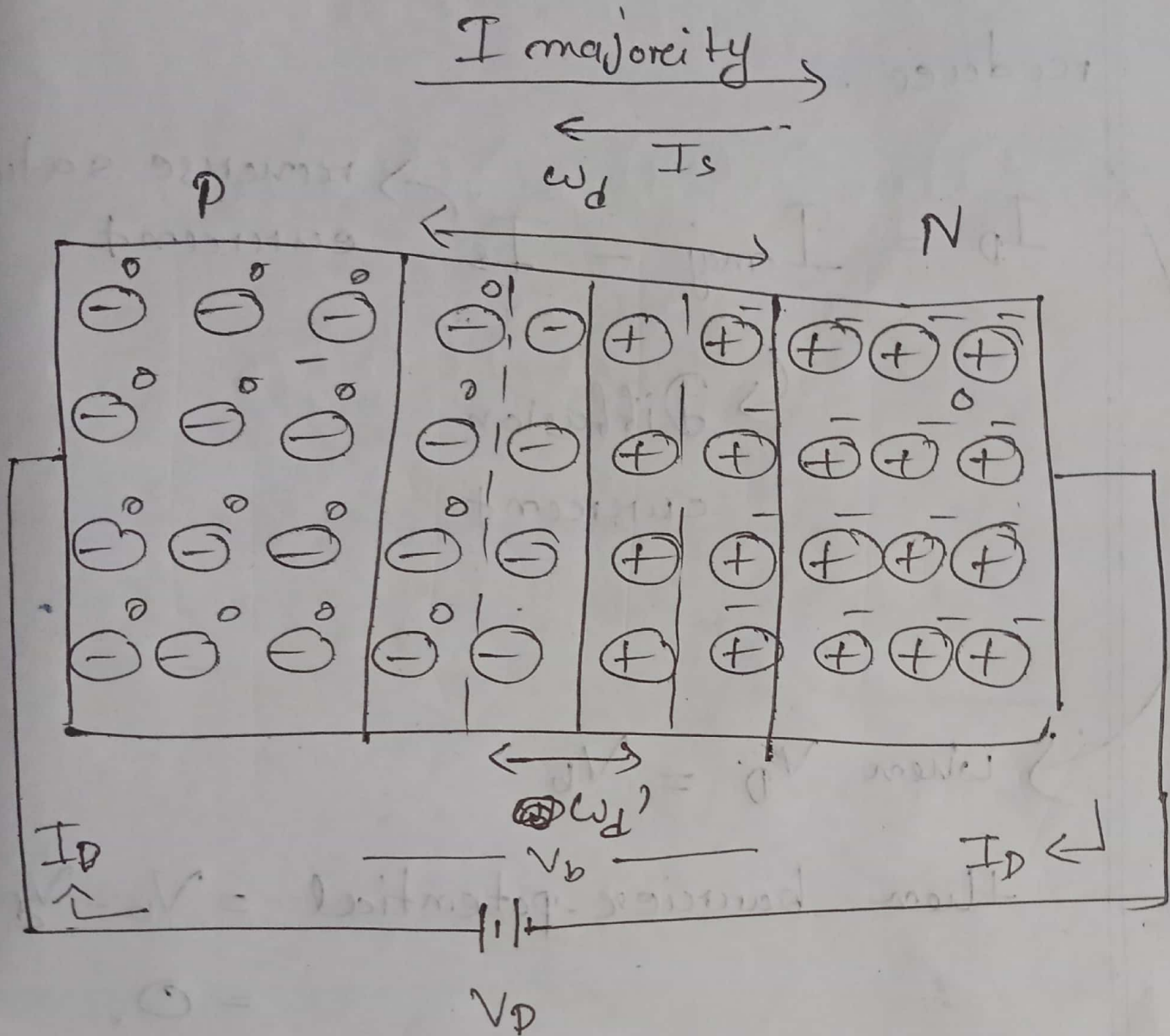


PN Junction Diode (Forward-Bias condition)



when $V_D = 0$,

width of depletion region is $= w_d$.

when $V_D > 0$

width of depletion region is $= w_d'$

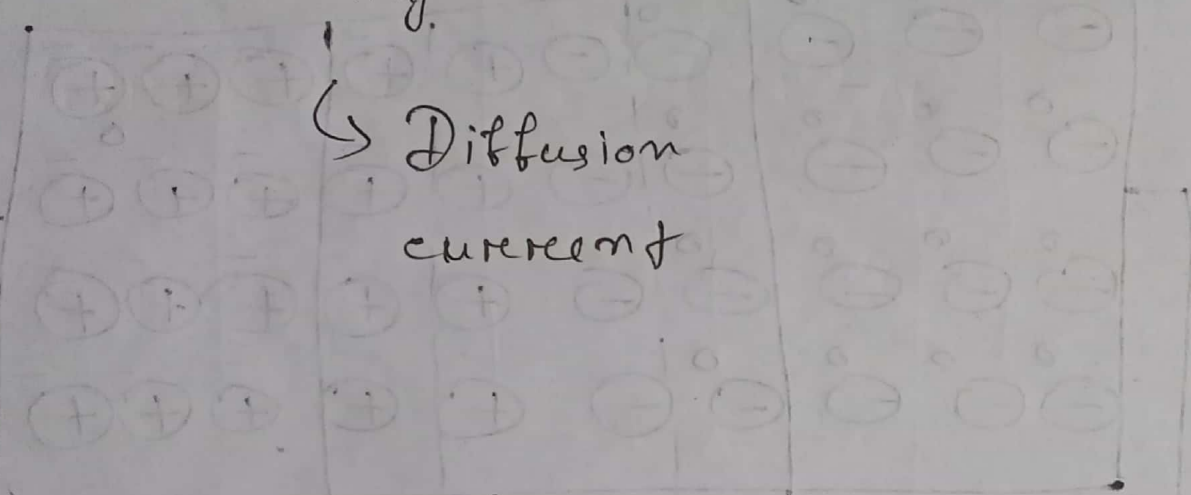
$w_d > w_d'$

As a result barrier potential will reduce.

$$I_D = I_{maj} - I_s$$

reverse saturation current

Diffusion current

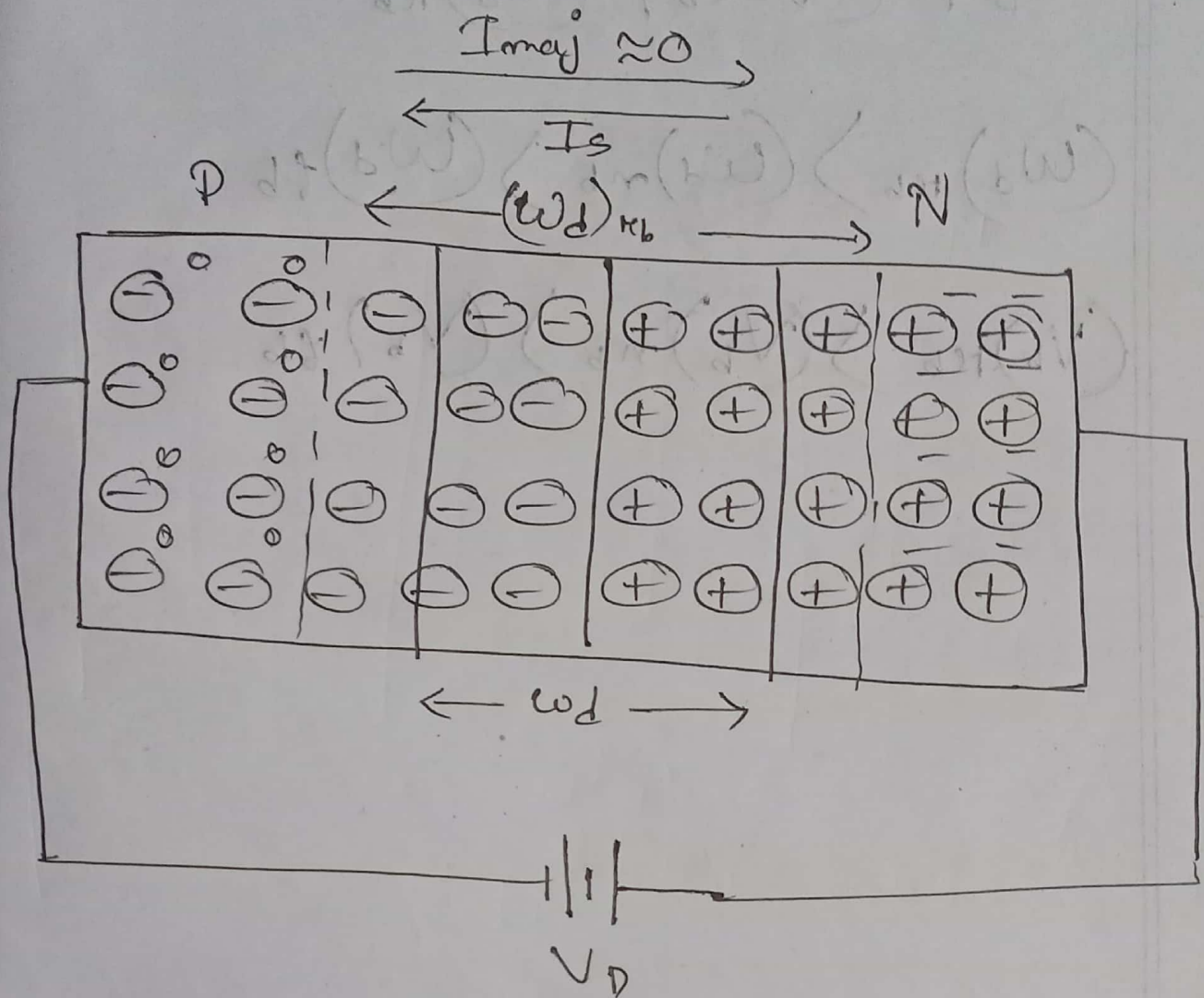


when $V_D = V_b$

then barrier potential $= V_b - V_D$
 $= 0$.

immense rise
in current
flow occurs

PN Junction Diode (Reverse Bias Condition)



in No bias condition:

width of depletion layer = W_d

barrier potential = V_b

in Reverse bias condition:

width of depletion layer = $(W_d)_{rb}$

barrier potential = $(V_b)_{rb} = V_b + V_D$

$$V_D \uparrow (V_b)_{KB} \uparrow (W_D)_{KB} \uparrow$$

$$(W_D)_{KB} > (W_D)_{MB} > (W_D)_{LB}$$

$$(V_b)_{KB} > (V_b)_{MB} > (V_b)_{LB}$$

+	+	+	+	+	-	-	-	-	-
+	+	+	+	+	-	-	-	-	-
+	+	+	+	+	-	-	-	-	-

← ω →

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V_D